

SUMMARY TABLE: PHYSICS GRADE 10

Sub-Strand 1.4: Energy, Work, Power and Machines — Teacher Reference (Pre-filled)

SUMMARY TABLE: PHYSICS GRADE 10	
Sub-Strand	Sub-Strand 1.4: Energy, Work, Power and Machines
Driving Question	DRIVING QUESTION: Why can a jua kali mechanic in Nairobi loosen a wheel nut effortlessly with a long spanner when a short one fails — and where does the energy go when friction makes the nut hot?

INSTRUCTIONS

FOR TEACHERS: This is the teacher reference version — each row is pre-filled with expected student responses. Use it to assess student Summary Tables, identify gaps, and prepare discussion questions. The student version has blank cells for students to complete after each lesson.

Lesson # and Title	What did I observe?	What did I learn?	How does this explain the phenomenon?
Lesson 1 Kinetic Energy and Gravitational Potential Energy: What Makes Things Move and Fall?	A jua kali mechanic in Gikomba, Nairobi, loosening a stubborn matatu wheel nut effortlessly with a long spanner-pipe extension after failing with a short spanner; a wheel nut that feels warm after repeated failed attempts; a Kenyan sprinter running at full speed; a stone falling from a grain-store roof in Kitale; PhET Energy Skate Park simulation showing energy bars changing with mass, speed, and height; a Kisumu construction-site video showing a stone making deeper dents when dropped from greater heights.	Kinetic energy (KE) is the energy an object has due to its motion: $KE = \frac{1}{2}mv^2$. Gravitational potential energy (PE) is the energy stored in an object due to its height above a reference point: $PE = mgh$. KE increases with both mass and the square of speed — doubling speed quadruples KE. PE increases linearly with both mass and height. Both are measured in Joules (J). In the mechanic's spanner system, the raised spanner arm stores PE and converts it to KE as it swings down. The heat in the wheel nut represents energy that has gone somewhere — but we cannot yet fully explain where.	Using $KE = \frac{1}{2}mv^2$, a 70 kg matatu tout running at 4 m/s has $KE = \frac{1}{2} \times 70 \times 4^2 = 560$ J. Using $PE = mgh$, a 5 kg bag of maize lifted 2 m onto a grain-store shelf in Kitale has $PE = 5 \times 10 \times 2 = 100$ J. The spanner arm before the mechanic's push stores gravitational PE (high position); as it swings, this converts to KE. The warm wheel nut is evidence that some energy is transformed into thermal energy — a mystery that points directly to the unit's driving question about energy conservation and dissipation, which will be fully resolved in subsequent lessons.
Lesson 2 Conservation of Mechanical Energy: The Invisible Transfer	Using the PhET Pendulum Lab simulation, learners observed that as a pendulum swings, kinetic energy (KE) is highest at the lowest point and potential energy (PE) is highest at the two extreme ends — and that the SUM of KE + PE remained approximately constant across all positions when friction was set to zero. When friction was introduced, the total energy SUM decreased over time. Learners also	Learners learned that kinetic energy ($KE = \frac{1}{2}mv^2$) is the energy of motion and potential energy ($PE = mgh$) is stored energy due to position. In a frictionless system, KE and PE continuously interchange but their total — mechanical energy — is conserved. This is expressed as $KE_1 + PE_1 = KE_2 + PE_2$. When friction is present, the total mechanical energy decreases — raising the new question of where that energy goes.	Learners applied the conservation of mechanical energy to the jua kali mechanic phenomenon: when the mechanic raises his arm (and the long spanner) before pushing, he stores potential energy in the system. As the spanner swings downward, that PE converts to KE — delivering force to the wheel nut. The class energy transformation model diagram shows this flow. However, the wheel nut getting hot remains

	annotated a roller-coaster diagram at five points, identifying where KE and PE were highest.		unexplained by conservation of mechanical energy alone, anchoring the next investigation into friction and thermal energy.
Lesson 3 Work: Definition and Calculation	Students pushed a school desk across the classroom floor, held a heavy schoolbag while walking horizontally, and resisted a desk push to simulate friction. They also observed data from the jua kali spanner scenario (short handle: $F = 250\text{ N}$, $d = 0\text{ m}$; long handle: $F = 80\text{ N}$, $d = 0.45\text{ m}$) and recorded force, displacement, and angle for each scenario.	Work is defined as $W = Fd \cos\theta$ — the product of force, displacement, and the cosine of the angle between them. When the angle is 0° , maximum positive work is done (force and motion in the same direction). When $\theta = 90^\circ$, work done is zero (e.g., carrying a bag horizontally). When $\theta = 180^\circ$, work is negative (friction opposes motion). The mechanic did zero scientific work with the short spanner because the nut did not move ($d = 0$), but 36 J of work with the long spanner. Friction converts mechanical energy to thermal energy — this is why the nut got warm.	The jua kali mechanic's short spanner failed not because of insufficient force alone, but because the nut's displacement was zero — so no work was done on the nut regardless of force applied. The mechanic's biological energy was still consumed, but converted to heat by friction at the nut-spanner interface rather than to useful rotational work. With the long spanner, a smaller force over a greater displacement arc achieved the required work (36 J) to overcome the nut's resistance — illustrating that work depends on both force AND displacement in the direction of the force.
Lesson 4 Work Done Against Friction: Where Does the Energy Go?	We pulled a wooden block across a smooth tile and a sandpaper surface using a spring balance and measured the friction force. We calculated the work done against friction using $W_f = f_r \times d$. After ten pulls on sandpaper, the surface felt noticeably warmer than after pulls on the smooth tile.	Work done against friction converts mechanical energy into thermal energy. The formula is $W_f = f_r \times d$ where f_r is the friction force in newtons and d is the distance slid in metres. Total energy is conserved — the thermal energy produced equals the mechanical energy input that went against friction. Rougher surfaces produce larger friction forces and therefore more thermal energy for the same distance.	The wheel nut at Gikomba felt warm after the mechanic pushed hard with the short spanner because even though the nut did not move (zero useful work output), there was still microscopic relative movement and deformation at the contact surfaces between the spanner and nut, and between the nut and bolt thread, converting the mechanic's muscular energy into thermal energy through friction. This is why the nut gets hot without tightening or loosening — the energy was not lost but transformed irreversibly into heat.
Lesson 5 Power: Definition, Calculation, and Everyday Applications	Watched a stair-climbing YouTube demonstration showing a student (Achieng) and a teacher (Mr. Kariuki) climbing the same staircase at different speeds, and collected mass, height, and time data into an observation table. Reviewed data from Eliud Kipchoge's sub-2-hour marathon attempt and the Mwea Irrigation Scheme pump specifications.	Power is the rate of doing work: $P = W/t$, measured in Watts ($W = \text{J/s}$). Doing the same work in less time requires more power. Power can also be calculated as $P = Fv$. A more powerful person or machine does not necessarily do more total work — they do the same work faster. The jua kali mechanic's speed of loosening the nut determines his power output; the long spanner determined his mechanical advantage. Friction heating the wheel nut is also an energy transfer happening at a rate.	Two people doing the same amount of work are NOT equally powerful if they take different amounts of time. Power measures HOW FAST work is done, not HOW MUCH work is done. Eliud Kipchoge outputs roughly $1,000\text{ W}$ (1 kW) during peak marathon running — comparable to a household appliance. The Mwea irrigation pump's power rating can be calculated from how fast it moves water against gravity. A more powerful machine does the same work faster, but this does NOT mean it is more efficient — efficiency is about how much of the input energy becomes useful

			output energy, which is our next investigation.
Lesson 6 Simple Machines: Types and Mechanical Advantage	Learners rotated through six lab stations representing Kenyan simple machines — a jua kali lever, a Kisumu harbour pulley, a Kericho tea farm inclined ramp, a borehole hand-pump wheel and axle, a sugarcane screw press, and a panga wedge. At each station, they used spring balances and physical models to measure the effort required to move a known load, recording data in a Comparative Data Table.	Learners calculated Mechanical Advantage ($MA = \text{Load} \div \text{Effort}$) for all six machine types and discovered that every machine with $MA > 1$ requires the effort to move through a greater distance than the load — confirming that $\text{Work In} = \text{Work Out}$ (ideally) and that no machine creates extra energy. They connected this directly to the Gikomba phenomenon: the long spanner extension increases the lever's effort arm, raising MA and reducing the effort needed to loosen the nut.	Using Claim–Evidence–Reasoning writing and a Jigsaw discussion, learners constructed the explanation that simple machines redistribute force and distance — trading one for the other while conserving total work done. They annotated cumulative models with a labelled lever diagram of the spanner system (fulcrum at the nut, effort arm = spanner + pipe extension, MA from measured data) and identified the friction point at the nut-bolt interface as the site of heat production — flagging this as the unresolved question for Lesson 7 on efficiency and energy dissipation.
Lesson 7 Velocity Ratio, Efficiency, and the Cost of Friction	Learners measured effort distance and load distance on a lever model at two different fulcrum positions, observed efficiency data for four different machines (lever, single pulley, wheelbarrow, screw jack) displayed in a gallery walk, and watched a video clip showing a pulley system at a posho mill near Eldoret to calculate VR and efficiency independently.	Velocity Ratio (VR) = effort distance $\div$ load distance; Efficiency (%) = $(MA \div VR) \times 100\%$; no real machine achieves 100% efficiency because friction always converts some work input into heat; machines redistribute force and distance but cannot create energy — the Law of Conservation of Energy is preserved and the warm wheel nut from Lesson 1 is the physical evidence of this energy conversion.	The jua kali mechanic's long spanner extension increases the effort distance relative to the load distance, giving a high VR and high MA — so less muscular force is needed. However, friction in the wheel nut threads and spanner joints converts some of the work input into heat (the warm nut observed in Lesson 1), meaning the useful work output is always less than the total work input. This is why efficiency is always below 100% and why no machine can produce more energy than is put into it.
Lesson 8 Synthesis, Model Finalisation, and DQB Resolution: The Full Story of the Jua Kali Mechanic	We observed that a jua kali mechanic in Gikomba, Nairobi, failed to loosen a wheel nut with a short spanner but succeeded effortlessly when he extended the spanner arm with a metal pipe. We also observed that the wheel nut became noticeably warm after repeated failed attempts with the short spanner, even though the nut did not move.	We learned that: (1) a longer effort arm increases mechanical advantage ($MA = \text{Load} \div \text{Effort}$), allowing a smaller force to produce a larger turning effect on the nut; (2) the total work input equals useful work output plus work done against friction ($W_{\text{in}} = W_{\text{out}} + W_{\text{friction}}$), so the machine does not create energy — it redistributes force and distance; (3) friction converts mechanical energy into thermal energy, which is why the nut gets hot; (4) efficiency ($\eta = MA/VR \times 100\%$) is always less than 100% in real machines because of friction; and (5) power ($P = W/t$) measures how quickly the mechanic does work — a stronger or faster mechanic has greater power output.	We can now fully explain the anchoring phenomenon: the long spanner creates a greater mechanical advantage by increasing the effort arm length, meaning the mechanic applies a smaller force over a larger distance to produce the same work (turning the nut). The machine does not create extra energy — it trades force for distance. The heat in the nut is explained by the work-energy principle: work done against friction is transformed into thermal energy, which dissipates into the metal. Because of this friction loss, the mechanic must always put in more energy than he gets out as useful work — no machine can exceed 100% efficiency, and energy is always conserved across the whole system.

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